

Mapping Financial Mindsets: A Two-Stage Unsupervised Framework for Behavioral Profiling Using High-Dimensional Psychometric Data

Irina Rabinovich¹, Rikki Rabinovich², Nick Ashburn^{1,3}, Moraya DeGeare^{1,4}

Abstract

Financial well-being reflects a complex interplay of psychological traits, behaviors, and knowledge, yet most segmentation approaches rely on limited indicators such as spending patterns or socio-demographic data, overlooking behavioral and psychometric factors. This paper proposes a novel methodological framework for psychometric segmentation that models complex interactions across multiple psychometric domains, enabling the identification of latent profiles of financial mindset and behavior. We demonstrate the utility of the framework by applying it to two distinct datasets--a proprietary financial assessment (N = 337) and the nationally representative U.S. CFPB Financial Well-Being Survey (N = 5,897). In both applications, the approach yields psychologically interpretable profiles, exhibits strong internal stability, and shows consistent associations with life satisfaction, psychological well-being, and financial health. Rather than prescribing a single population taxonomy, our goal is to provide a replicable and scalable procedure for identifying meaningful behavioral patterns within any financial population. This makes the approach well suited to behavioral finance applications including financial advising, wellness interventions, and the design of personalized tools and communications aimed at improving financial decision-making and well-being.

1. Introduction

Financial well-being is increasingly recognized as a multidimensional construct that includes not only objective indicators, such as income, savings, and debt, but also subjective experiences,

¹ Calibrate, Inc; ² Department of Neurosurgery, University of Utah; ³ Drexel University; ⁴ BFF Therapy

including financial stress, confidence, and perceived control (Brüggen et al., 2017; CFPB, 2015a; Netemeyer et al., 2018). It reflects both behavioral and psychological aspects, demanding assessment tools that go beyond financial conditions to account for beliefs, motivations, and emotional responses. Traditional methods for classifying individuals in the financial domain have typically relied on spending patterns or demographic factors and have therefore often overlooked the psychological patterns that influence how individuals perceive, relate to, and manage their finances. Here, we define financial mindset as a multidimensional construct where capability, motivation, psychological resources, and enacted behavior intersect to shape decision-making. This perspective aligns with our clustering framework, which seeks to identify patterns in how these dimensions co-occur across individuals. The resulting clusters represent distinct profiles of how individuals organize and regulate their financial lives.

This paper makes three contributions. First, it addresses the challenge of modeling multifaceted psychological constructs and their interdependencies in behavioral profiling. To this end, it introduces a methodological framework that integrates domain-specific manifold learning with spectral clustering to identify nonlinear patterns in psychometric and behavioral data. Second, it tests the framework across two distinct datasets—one proprietary and psychometrically rich, the other large-scale and nationally representative -- demonstrating how this framework reveals psychologically coherent patterns of financial mindsets that are not captured by conventional linear or demographic segmentation approaches. Third, as the results of the application to the two distinct datasets show, it provides preliminary evidence that the proposed approach reveals a reliable underlying structure – showing consistent separability and stability across different instruments, populations, and survey formats, even if the specific profiles differ.

Applying this two-stage unsupervised learning framework, we identify emergent

behavioral profiles that capture complex nonlinear relationships missed by standard factor-analytic or demographic approaches. We evaluate these profiles internally using structural metrics of cluster coherence and externally by demonstrating robust associations with outcome indicators, namely, life satisfaction, psychological well-being, and financial health. We also show that these external indicators are far more strongly associated with the learned profiles than with socio-demographic variables such as age, education, or income, underscoring the distinctive value of psychometric and behavioral dimensions for understanding heterogeneity in financial well-being. Finally, we show that demographic variables alone provide only limited predictive power for cluster membership, indicating that the framework captures behavioral–psychological structure that is not reducible to age, income, or education.

To demonstrate the utility of the proposed framework, we apply it to two very different datasets: a proprietary assessment with deep psychometric coverage and the U.S. CFPB Financial Well-Being Survey, which uses a different item structure and response format. Despite these differences, the approach consistently identifies interpretable, psychologically coherent profiles that share a common structural organization and exhibit strong internal structure and meaningful associations with external outcomes.

By decoupling the modeling methodology from any specific survey instrument, the framework opens the door to scalable application for academic research, personalized fintech solutions, and even clinical applications where financial behavior may inform intervention strategies.

2. Background & Related Work

In this section we review core concepts from behavioral finance and psychometrics that underpin our framework. We also highlight relevant applications of machine learning in the behavioral

sciences and examine prior attempts to assess financial well-being through multidimensional measurement.

2.1 Behavioral Finance and Psychometrics

Behavioral finance integrates psychological theories into economic models to better explain real-world financial decision-making, recognizing that individuals are often influenced by cognitive biases, emotions, and heuristics rather than strict rationality (Kahneman & Tversky, 1979). Psychometric approaches complement this perspective by using validated instruments to quantify individual differences in traits such as risk tolerance, financial anxiety, and internalized money beliefs or “scripts” (Grable & Joo, 2004; Klontz et al., 2011; Lown, 2011).

A growing body of evidence shows that psychological traits, like self-control, optimism, and emotional regulation, are strongly associated with financial behaviors (e.g., saving, spending, and investing) and contribute to broader measures of well-being (Gathergood, 2012; Shim et al., 2009; Xiao & O’Neill, 2016). Perceived psychological power has also been shown to promote financial behaviors like saving, highlighting the affective and motivational dimensions of financial decision-making (Garbinsky et al., 2014). The integration of psychometrics into financial assessments improves the predictive accuracy of financial health outcomes by capturing psychological contributors beyond financial variables alone (Serido et al., 2013). Building on this foundation, our work fuses self-reported financial behavior constructs with multidimensional psychological traits into a unified representation, enabling more precise behavioral profiling than isolated domain-specific approaches.

2.2 Clustering and Dimensionality Reduction in Behavioral Sciences

Unsupervised learning methods have become increasingly valuable in the behavioral sciences, enabling researchers to uncover natural groupings within complex behavioral datasets (Kaufman

& Rousseeuw, 2009; Yarkoni & Westfall, 2017; Zhang, 2021). In finance, these techniques have primarily been applied to behavioral segmentation based on observable financial actions, credit risk modeling, and product targeting (Nazareth, 2023). Dimensionality reduction methods such as Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP¹) aid in simplifying high-dimensional datasets while preserving structure relevant to clustering (Jolliffe, 2002; McInnes et al., 2018). Combinations of nonlinear embedding and graph-based clustering have been successfully applied in other domains, including genomics (Becht et al., 2019), psychiatry (Marquand et al., 2016), and health informatics (Miotto et al., 2016), where they have enabled the discovery of complex latent structure that is difficult to recover with linear or centroid-based methods. However, applications of these approaches to financial–psychometric segmentation remain limited.

Unsupervised learning is particularly appealing in contexts where ground-truth labels are unavailable or poorly defined, allowing researchers to detect latent patterns inherent in the data. In the context of our study, these methods can reveal psychologically meaningful profiles, especially when working with rich psychometric or behavioral data that is difficult to analyze directly. However, critiques of unsupervised methods note their sensitivity to parameter choices, potential instability across different samples, and challenges in validating clusters against external criteria (Şenbabaoğlu et al., 2014; von Luxburg, 2010). We address these concerns through internal validation metrics, external outcome associations, and replication across independent samples with different survey structures.

Recent studies have successfully applied unsupervised methods to reveal latent structure in

¹ A dimensionality reduction technique that transforms high-dimensional data (e.g., multiple questionnaire scores) into a lower-dimensional representation while preserving underlying structure. It facilitates pattern recognition and visualization.

other (not financial) psychological domains. For example, Hofman et al. (2023) used hierarchical clustering to identify stable subtypes of depression and anxiety symptoms in a large-scale population study, while Möttus et al. (2017) demonstrated the predictive utility of fine-grained personality trait clustering for life outcomes. These studies highlight the potential of clustering to uncover psychologically meaningful profiles that support both explanatory modeling and tailored interventions. We go beyond these innovations and apply UMAP and Spectral Clustering to integrated psychometric and self-reported financial behavior data, generating psychologically interpretable behavioral-financial profiles grounded in latent structure.

2.3 Prior Approaches to Financial Well-Being Assessment

While traditional assessments of financial well-being have evolved to include subjective indicators such as financial stress, perceived control, and self-efficacy (CFPB, 2015b; Netemeyer et al., 2018), they remain limited in psychological scope. Tools like the CFPB Financial Well-Being Scale and the InCharge Distress Scale (Prawitz et al., 2006), though psychometrically strong ($\alpha \approx 0.87$), primarily assess emotional strain and perceived hardship. They omit motivational drivers, behavioral consistency, and self-regulatory capacity. Similarly, financial literacy assessments such as the OECD/INFE Toolkit (Atkinson & Messy, 2012) emphasize knowledge but not emotional readiness to act. This is a critical omission, since traits like impulsivity, perseverance, and low self-efficacy often inhibit behavior change, regardless of knowledge.

Psychological constructs such as locus of control, psychological capital, and financial self-efficacy have been linked to financial behaviors (Grable & Joo, 2004; Lown, 2011; Serido et al., 2013), but they are typically modeled in isolation or examined using supervised methods. This obscures how multiple traits co-occur in practice and limits the discovery of latent

psychological configurations. Machine learning in fintech, while increasingly common (Das & Nayak, 2022; Zhang, 2021), prioritizes what users do (observable output) over why they do it (internal psychological structure), limiting the ability to predict how financial stress will impact decision-making.

The present study addresses these limitations by clustering a comprehensive set of psychometric and behavioral measures to reveal interpretable financial well-being profiles. By modeling the co-occurrence structure across domains, the framework uncovers psychologically meaningful patterns that are difficult to detect using linear or domain-isolated approaches. These profiles do not represent fixed “types” but rather data-driven configurations that capture how traits, behaviors, and motivations align within and across individuals. This structure enables richer descriptive insight, supports hypothesis generation for future theory-building, and can inform targeted interventions in financial advising and wellness contexts.

3. Framework Overview

The framework begins by organizing questionnaire items into conceptually coherent domains, each representing a distinct psychological or behavioral construct (e.g., motivation, planning, financial literacy). Domain partitioning is essential because psychometric surveys typically contain a heterogeneous mix of measurement formats--binary indicators, multi-level Likert items, and scales with different response ranges. Even when items are rescaled to a common numeric range, the combined item space remains geometrically inconsistent: binary items create abrupt, discontinuous jumps in response patterns; Likert items introduce smooth graded variation; and items from unrelated constructs add uncorrelated or orthogonal sources of variance. When these heterogeneous items are analyzed or clustered directly at the item level, the resulting structure is often dominated by measurement artifacts rather than underlying behavioral

organization, leading to clusters that change unpredictably across seeds or algorithms. Indeed, we are unable to obtain interpretable item-level clusters in either dataset used in this study, even after applying dimensionality reduction to raw responses. In contrast, partitioning the survey into domains preserves construct-level coherence: items designed to measure the same latent trait are modeled together, while variation across constructs is deferred to later stages. The domain structure provides the foundation for Stage 1 of the framework, which produces continuous, domain-level scores for each participant. These domain scores then serve as the standardized, psychometrically meaningful inputs to Stage 2, which identifies behavioral clusters based on the joint distribution of domains. Figure 1a provides an overview of the full two-stage dataflow.

3.1. Considerations of Domain Depth and Response Variability

To produce a meaningful domain-level representation, the domain must have sufficient *response resolution*, i.e., items within a domain must collectively span a sufficiently rich set of response patterns to capture meaningful distinctions among individuals. Domains with very few items or coarse response scales yield a limited set of possible response patterns, making it difficult to differentiate individuals or summarize the construct along a continuum. In practice, domains intended to reflect broad, multi-faceted constructs typically require enough item-level richness – at least six to eight items with nontrivial variability – to behave like continuous dimensions. By contrast, domains designed to measure narrow or tightly defined attributes (e.g., financial literacy, numeracy, construal level, or short-form affective or motivational mini-scales) naturally exhibit lower variability and fewer response combinations; these domains are scored using simpler approaches that reflect their focused scope. Domains with insufficient response resolution for continuous differentiation are excluded from the clustering stage but may be retained for interpretive or validation analyses. These considerations determine how each domain is handled

within the framework’s scoring procedures introduced below.

3.2 Stage 1: Producing Unidimensional Domain Scores

There are multiple valid ways to construct a domain-level score once items have been grouped into coherent constructs, and the appropriate approach depends on the statistical and psychometric properties of the domain. In this paper, we explore three such strategies, chosen to illustrate how the framework can adapt domain scoring to different structural patterns. Other scoring approaches--such as projections in the original item space or alternative aggregation schemes--are also feasible, but evaluating all possible variants lies beyond the scope of the present study.

3.2.1 Anchor Projection Score

For medium- to high-dimensional domains with nonlinear relationships between the items, heterogeneous item formats, and high response resolution, we apply dimensionality reduction followed by *anchor-based projection*. By embedding high-dimensional response patterns into a lower-dimensional space, dimensionality reduction can mitigate redundancy, noise, and uneven scaling, thereby improving interpretability and support downstream clustering. In principle, both linear methods, such as principal component analysis (PCA), and nonlinear manifold learning techniques, such as UMAP, can be used. Linear methods assume that domain structure is well approximated by a global linear subspace and are effective when item relationships are predominantly linear. Nonlinear methods relax this assumption by allowing local neighborhoods to be preserved even when the global structure is curved. Note that some nonlinear approaches, such as t-SNE, are primarily optimized for visualization and do not preserve global geometry or support stable one-dimensional parametrization, making them less suitable for domain scoring in the present context.

To guide method selection, we assess the geometric structure of each domain using several complementary diagnostics. One indicator is the variance explained by the first principal component, which reflects the dominance of a single linear dimension. We also quantify nonlinearity using a tangent-alignment index, defined as the mean absolute cosine similarity between the global PC1 direction and local one-dimensional PCA tangents estimated from each participant's 20 nearest neighbors. Domains with high PC1 dominance and tangent alignment near 1.0 behave approximately linearly, whereas lower values indicate curved latent structure. As an additional check, we evaluate whether linear projections adequately preserve neighborhood relationships using trustworthiness and continuity metrics.

Once a domain representation is obtained, we convert each participant's multidimensional coordinates into a unidimensional domain score using a deterministic *anchor-based projection*. For each domain, we define two deterministic anchor profiles: one representing the weakest possible response for the domain traits, $\mathbf{v}_{min}$ (e.g., all 1s), and one the strongest, $\mathbf{v}_{max}$ (e.g., all 5s). These anchors are embedded into the same low-dimensional space as the observed participants, yielding embedded anchor points $\mathbf{v}_{min}^e$ and $\mathbf{v}_{max}^e$ respectively. Because UMAP embeddings typically organize observations along smooth latent gradients, projecting participants onto the axis defined by the embedded anchors provides a simple and interpretable summary of the dominant pattern distinguishing individuals within a domain. When this latent gradient is approximately linear, linear projection preserves both relative ordering and distances along the principal axis of variation. In this case, the domain score for participant i is computed as the orthogonal projection of the participant's embedding $\mathbf{p}_i$ onto the anchor axis: $s_i = [(\mathbf{p}_i - \mathbf{v}_{min}^e) \cdot (\mathbf{v}_{max}^e - \mathbf{v}_{min}^e)] / \|\mathbf{v}_{max}^e - \mathbf{v}_{min}^e\|^2$. This score represents the participant's relative position along the axis spanning the theoretical response range.

However, when the gradient is substantially curved, for instance, due to saturation effects, linear projection can distort distances or relative ordering along the manifold. To assess whether linear projection is adequate, we evaluate rank consistency between anchor-based linear scores and manifold-respecting positions using Spearman’s rank correlation coefficient (ρ) and Kendall’s rank correlation coefficient (τ). These metrics quantify the degree to which linear projection preserves relative ordering along the underlying manifold. Linear projection is retained when $\rho \geq 0.90$ and $\tau \geq 0.85$. When both statistics fall below these thresholds, indicating concurrent global and local rank distortion, we compute participants’ scores using geodesic distances (e.g., via Isomap) measured along the embedded manifold from $\mathbf{v}_{min}^e$ to each participant, normalized by the total manifold path length connecting $\mathbf{v}_{min}^e$ and $\mathbf{v}_{max}^e$ (Figure 2). Although geodesic scoring provides a manifold-respecting parametrization of domain progression, it is not used by default: if the latent gradient is approximately linear, linear anchor-based projection yields more stable and auditable scores. Geodesic scoring is therefore applied only when diagnostics indicate that linear projection fails to preserve both global and local rank structure.

Unlike sample-based scores derived from z-scoring, percentile ranking, or other normalization schemes, where scores depend on a participant’s relative position within a given sample, anchor-based projection retains absolute interpretive meaning by computing each score relative to fixed conceptual anchor profiles rather than to other respondents in the dataset. Sample-based scores are affected by the empirical distribution of the data, which can distort interpretation and make respondents appear “low” or “high” simply because the sample is skewed. In addition to making scores sample-independent, anchor-based projection offers a

preferable alternative to simple and weighted averages in domains where appropriate item weights are unknown or direct aggregation of item values is not appropriate. By placing each respondent along a predefined continuum between low-functioning and high-functioning anchor profiles, this approach produces domain-level scores without requiring explicit weighting assumptions or reliance on sample-dependent normalization.

3.2.2 Weighted Average Score

When anchor-based projection is not applicable, such as in domains composed of dichotomously scored items (e.g., correct vs. incorrect) or domains with highly homogeneous items reflecting a particularly narrow trait, a weighted average provides a suitable alternative. In these cases, the construct is well defined in advance, item directionality is clear, and either the relative importance of items is known or can be reasonably derived. In these cases, a weighted composite score offers a stable and interpretable summary of each participant's standing in the domain. Below we describe two typical scenarios when weights can be derived.

If items function as indicators of a common latent construct (e.g., abstract vs. concrete thinking), item weights may be derived from the dominant covariance pattern within the domain. Indicators with stronger and more consistent contributions to shared variance receive higher weights, while weaker indicators contribute proportionally less. This yields a weighted index that retains the clarity of the composite score while incorporating information about the latent structure implied by the data. Weighting indicators by their contribution to shared variance is consistent with reflective measurement theory (e.g., Bollen, 1989; Fabrigar & Wegener, 2012; Embretson & Reise, 2000), which defines constructs through systematic covariance among indicators.

By contrast, in domains where responses indicate correctness rather than endorsement of

a psychological attribute, a difficulty-based weighting scheme can be applied, in which each item's weight is proportional to its empirical difficulty (based on the prevalence of correct responses in the sample) after adjusting for chance performance (e.g., Lord, 1989; Embretson & Reise, 2000). Harder items receive higher weights, whereas items answered correctly by nearly all participants contribute less.

After computing item weights, we compute domain scores as difficulty- or covariance-weighted proportions, with 0 corresponding to the lowest attainable (not necessarily present in the sample) standing and 1 to the highest. The resulting scores—bounded by the theoretical range of the corresponding trait rather than the range in the sample—places these domains on the common scale with the anchor-based projections described above, ensuring comparability across domains and compatibility with downstream analyses.

3.2.3 Simple (Normalized) Average Score

For domains that do not lend themselves to anchor projection or are composed of items for which no principled basis exists for differential weighting, we summarize responses using simple unweighted averages. This approach is appropriate when items share a common response scale, ensuring comparability across indicators². Under these conditions, equal weighting provides a transparent summary of domain-level functioning without imposing additional assumptions about item importance or latent structure.

3.3 Stage 2: Clustering

The second stage of the pipeline operates on the domain-level scores produced in Stage 1.

Because each domain is now represented by unidimensional scores and the number of domains is

² For domains that don't fall into any of the three categories we consider in this paper, alternative scoring mechanisms (such as anchor projection in the original space) will need to be considered. None of the domains we identified in the datasets used for the current study required that.

generally small, the input to stage 2 is therefore low-dimensional. Nonetheless, domains are intentionally constructed to be conceptually distinct and hence, when considered jointly, can exhibit nonlinear structure, heterogeneous variance, or uneven resolution. To accommodate these cross-domain characteristics, we apply UMAP to the domain-level scores to construct a second-stage embedding prior to clustering.

To show that nonlinear embedding is called for at this stage, we benchmark clustering performance across raw domain scores, PCA projections, and UMAP embeddings. UMAP produces markedly stronger separation and more coherent geometry than either PCA or raw-space representations, as reflected in higher silhouette values across benchmarked configurations (Appendix B). We expect this finding to hold generally if survey items are grouped into domains properly, i.e., to represent largely unrelated psychometric constructs.

Spectral clustering is then performed to identify latent groupings within the embedded space³. The resulting clusters of the points in the embedded space correspond to distinct psychometric profiles of the participants represented by those points. Both UMAP transformation and spectral clustering will produce different clustering solutions depending on hyperparameters used. We tune hyperparameters and evaluate resulting clustering solutions based on internal validity and qualitative interpretability of the resulting profiles. Internal validity is quantified using standard metrics--including the silhouette score, Calinski–Harabasz index, and Davies–Bouldin index--computed in the embedded space, while qualitative interpretability is assessed by the coherence and distinctiveness of domain score patterns across clusters. These complementary considerations guide selection among candidate solutions in specific applications

³ We tested alternatives such as K-means, DBSCAN, and HDBSCAN, but these performed poorly in capturing the non-convex geometry of the UMAP-transformed space (see Appendix B for comparisons).

of the framework.

We note that domains with few distinct score values (i.e., each value is shared by a significant fraction of participants) can induce trivial clustering partitions, even when those domains have high theoretical response resolution. Because UMAP prioritizes the strongest local contrasts when constructing its neighborhood graph, these low-cardinality domains produce a small number of discrete plateaus that dominate the remaining dimensions. Spectral clustering then splits the data almost exclusively along these plateaus, yielding clusters that are technically stable but largely uninformative: individuals with identical values on the low-cardinality domain are grouped together even when they differ substantially on the broader psychometric constructs. To avoid this collapse of the multivariate structure onto a single coarse dimension, we exclude these domains from Stage 2 clustering and retain them only for interpretive analyses.

To aid interpretability, clusters are indexed according to their average cross-domain behavioral standing. For each participant, we compute a scalar summary score defined as the mean of their standardized scores across the five behavioral-financial domains. This participant-level average provides a parsimonious measure of overall functioning that weights each domain equally and remains independent of geometric properties of the low-dimensional embedding. Cluster-level summaries are obtained by averaging these participant-level scores across all members of each cluster and clusters are then ordered according to their mean score, yielding an interpretable ordering along a continuous cross-domain gradient.

To aid interpretability, clusters are indexed according to their average cross-domain behavioral standing rather than their position in the embedded space. For each participant, we compute a scalar score defined as a weighted average of their domain scores, where weights reflect empirically derived domain importance from post hoc analyses of cluster separation.

Cluster-level summaries are obtained by averaging these participant-level scores across all members of each cluster. Clusters are then ordered according to their mean participant score, yielding an interpretable ranking along a continuous cross-domain gradient from lower to higher overall behavioral and financial functioning. This ordering is not intended to imply a normative ranking; rather, it provides a stable indexing convention across pipeline runs with different seeds and parameters. Nonetheless, the resulting numbering often follows a broad gradient--from more vulnerable or struggling profiles to more adaptive or flourishing ones. Intermediate clusters, however, do not fall cleanly along this continuum and instead reflect distinct, non-linear configurations of behavioral and psychological traits.

4. Use Case 1: Development of Individualized Strategy for Financial Advice

This framework was developed to support professional practice settings in which individuals or couples seek guidance around financial decision-making and money-related relationship dynamics. In such contexts, practitioners often work with clients who struggle with follow-through, self-regulation, emotional responses to money, or alignment between intentions and behavior, particularly among couples. The framework is designed to reveal a client's behavioral and psychological profile and thus support structured conversations, targeted coaching, and more tailored intervention strategies. This applied orientation motivates the use of a broad psychometric assessment of financial functioning that is given to new clients as part of onboarding. Responses to this assessment form a proprietary dataset which we describe next.

4.1 Dataset Description and Conceptual Coverage

The proprietary dataset consists of responses to a 122-item psychometric instrument, comprising 95 domain items that assess eight dimensions of financial functioning (Table 1) and 21 outcome items used exclusively for external validation analyses (described in Section 4.7). The instrument

draws on established frameworks including Family Financial Socialization Theory (Gudmunson & Danes, 2011), Self-Determination Theory (Di Domenico et al., 2022), and the Transtheoretical Model of Change (Prochaska & DiClemente, 1983).

The instrument includes eight domains, each containing between 5 and 21 items, selected to reflect multiple aspects of psychological functioning relevant to financial behavior, including self-regulation, motivation, construal, emotional resources, and financial knowledge. Domains corresponding to Financial Behavior (Prochaska & DiClemente, 1983), Financial Literacy (Lusardi & Mitchell, 2011), and Motivation (Ryan & Deci, 2017) are well-established predictors of financial well-being and provide familiar interpretability within behavioral finance. Attitudes & Capabilities domain captures perceived control, emotional tendencies, and financial self-concept—constructs strongly linked to planning and follow-through (Gudmunson & Danes, 2011) but often underrepresented in financial assessments. Psychological Capital (hope, resilience, optimism, efficacy; Luthans, et al, 2007) and Character Strengths (e.g., perseverance, self-regulation, gratitude) draw on well-validated frameworks (Peterson & Seligman, 2004) and have been empirically linked to financial resilience, adaptive coping, and reduced financial stress (Strömbäck et al., 2017; Xiao & O'Neill, 2016). Financial Construal Level, grounded in Construal Level Theory (Trope & Liberman, 2010), assesses how individuals construe and interpret financial choices, shaping risk perception, time horizons, and behavioral follow-through. The breadth of the item pool allows broad behavioral–psychological profiles to emerge during the unsupervised learning.

4.2 Participant Sample and Psychometric Properties

A total of 337 participants meet the predefined inclusion criteria and are included in the analytic sample. Participants are required to complete at least 80% of each survey module to ensure

sufficient item-level data for reliable domain score computation. Item-level missing responses within included surveys are handled using the imputation procedures described in Supplemental Table C3. Domain-level score distributions are examined for extreme values ($|z| > 3$), and no outliers are detected within the analytic sample. Participants are drawn from several U.S.-based sources—including financial advisors and their clients, college students, and voluntary users—resulting in a heterogeneous sample that spans a wide range of financial experiences, decision-making contexts, and behavioral tendencies. Table 3 summarizes the socio-demographic composition of the dataset.

Although modest in size, the dataset exhibits substantial response variability and strong internal reliability in all psychometric domains (Cronbach's $\alpha = .50-.94$; Supplemental Table S1). Responses vary widely across the full response range in each domain, with no evidence of floor or ceiling effects. Inter-domain correlations ranged from .30 to .83 (median $r = .56$), indicating related but non-redundant constructs. Prior methodological work in exploratory psychometric settings suggests that when measurement quality is high, meaningful latent structure can be recovered with modest sample sizes, even in non-probability samples (Kyriazos, 2018). Moreover, the sample composition reflects the kinds of mixed recruitment sources commonly available in practical behavioral-finance applications, where segmentation models are typically developed from convenience-based cohorts rather than nationally representative surveys.

4.3 Producing Domain Scores

Across the psychometric domains examined in this dataset, diagnostic analysis indicates that most domains exhibit sufficient response resolution and globally curved geometry (Table 2), warranting the application of UMAP followed by anchor-based projection to derive

unidimensional domain scores. To obtain stable embeddings, UMAP hyperparameters are selected via randomized search by jointly maximizing trustworthiness and continuity using a weighted objective to prioritize trustworthiness ($\alpha = 0.6$; see Appendix C for the resulting hyperparameter values). The embedding quality is high across all domains, with trustworthiness ranging from 0.936 to 0.958 and continuity ranging from 0.949 to 0.972. For all evaluated domains, linear anchor-based scores and geodesic (Isomap-based) scores produced nearly identical rank orderings, with Spearman correlations exceeding .99 and Kendall τ values above .87. Anchor-based projection scoring is, therefore, used throughout.

Two domains require alternative scoring approaches. Financial Literacy is scored as a difficulty-adjusted accuracy composite, reflecting the proportion of correct responses weighted by item difficulty as described in Section 3.2.2. In the Financial Construal Level domain, although items are binary, their response options differ substantially in the degree to which they reflect abstract versus concrete construal. Treating all responses as uniformly equivalent (e.g., assigning 0/1 values) would therefore obscure meaningful differences in abstraction across items. Instead, we use covariance-weighted aggregation, allowing responses to given items to contribute proportionally based on their empirical alignment with overall construal patterns in the sample. For example, items for which the “abstract” response tends to co-occur with abstract responses on other items receive greater weight.

4.4 Clustering and Internal Validation

Using the domain-level scores described above, we construct a second-stage UMAP embedding and apply spectral clustering to identify latent behavioral profiles of the participants. We tune the hyperparameters of both UMAP and spectral clustering (Appendix C) to optimize separation in the resulting low-dimensional manifold. Grouping the participants into seven clusters produces

results with discernible spatial separation and coherent structure: Silhouette Score of 0.569, Calinski–Harabasz Index of 1,543.50, and Davies–Bouldin Index of 0.579. Moreover, the clusters have reasonably uniform distribution as shown in Table 4.

To assess stability, we rerun the (UMAP → spectral clustering) pipeline using 100 independent random seeds while holding all hyperparameters fixed. Cluster consistency across runs is high (mean ARI = 0.916, SD = 0.043). We further assess robustness of the clustering stage to sampling variation: we generate a single UMAP embedding using the full dataset and apply spectral clustering to 100 random 80% subsamples within this fixed embedded space. Subsample solutions show very high stability relative to the full-sample reference (ARI = 0.970, SD = 0.018).

4.5 Cluster Interpretability

In addition to the internal validity as described above, the clusters also possess coherent and interpretable psychological structure (Figure 3a). These clusters can be understood as distinct profiles within the broader financial mindset space defined in the Introduction. They are defined not only by the overall level of functioning but by how distinct combinations of financial behavior, perceived capability, motivation, psychological resources, and indicators of positive functioning (defined by domain scores) combine to shape behavior. Although clusters are broadly ordered along a dominant functioning gradient, interpretability arises from meaningful deviations from this gradient, yielding distinct behavioral pathways rather than simple severity tiers.

At lower levels of functioning, limited psychological resources and motivation constrain the translation of financial knowledge into consistent behavior, whereas mid-range clusters diverge in whether financial engagement is primarily cognitive (e.g., higher construal and literacy

without equivalent execution) or structural (e.g., stable routines and competence with modest motivational drive). At higher levels of functioning, clusters further differentiate based on the role of motivation relative to skills, distinguishing profiles characterized by steady, competence-driven performance from those marked by heightened motivational activation and behavioral intensity. These patterns indicate that clusters reflect qualitatively distinct configurations of psychological and behavioral factors, rather than points along a single continuum. Domain-level summaries for cluster profiles are reported in Table 5a.

4.6 Demographic Analysis

To evaluate whether the behavioral clusters are primarily attributable to socio-demographic stratification, we model cluster membership as a function of age, education, and income using multinomial logistic regression. These variables are widely used indicators of socioeconomic status (SES) in behavioral and financial research. The overall model is statistically significant (likelihood-ratio test, $p < .001$), indicating that demographic variables are associated with cluster assignment. However, explanatory power is modest (McFadden pseudo- $R^2 = .091$), suggesting that socio-demographic factors account for only a limited proportion of variance in cluster membership. At the predictor level, income shows statistically significant associations with membership in two clusters ($ps = .002-.003$), while its associations with other clusters are non-significant ($ps > .08$). Age and education display isolated significant contrasts ($ps \approx .03-.05$), but the majority of their coefficients are non-significant ($ps \geq .09$), and no consistent pattern emerges across clusters. Taken together, these findings indicate that although demographic variables contribute some signal, they do not substantially account for the clustering structure.

4.7 External Associations with Independent Outcomes

We have seen in Sections 4.4 and 4.6 that our clustering approach expresses meaningful internal structure that cannot be revealed through purely demographic data. We now turn to another important aspect in evaluating our framework, namely, whether the latent profiles it reveals correspond to meaningful differences in overall functioning across psychological and behavioral domains. To this end, we consider how cluster membership correlates with independent external outcomes that were not utilized in cluster construction.

Specifically, we consider life satisfaction, psychological well-being, and perceived financial health – conceptually related yet empirically distinct evaluative outcomes. Life satisfaction reflects a global cognitive appraisal of one’s life circumstances, psychological well-being captures eudaimonic functioning and internal psychological resources, and financial health reflects perceived or objective financial capacity and security. Prior research consistently shows that although these constructs are moderately correlated, none is reducible to the others, and each retains substantial unique variance after controlling for socioeconomic status and objective financial conditions (Mentus, 2017; Kokun et al., 2024; Halandová, 2024). Moreover, psychological well-being is not merely an outcome of financial conditions but also a key antecedent of financial behavior: low psychological resources, maladaptive beliefs, and emotional distress reliably inhibit planning, follow-through, and adaptive financial decision-making, contributing to poorer financial health over time (Thompson, 2021; Thompson, 2023). Life satisfaction is measured using the Satisfaction with Life Scale (Diener et al., 1985), psychological well-being is assessed with the Flourishing Scale (Diener et al., 2010), and perceived financial health is evaluated using a 10-item version of the Financial Health Network Score (Financial Health Network, 2018).

Cluster membership is strongly associated with all three outcomes. One-way ANOVAs

indicate significant between-cluster differences for Life Satisfaction, $F(6,330) = 12.98$, $p < .001$ ($\eta^2 = .19$), Psychological Well-Being, $F(6, 330) = 19.90$, $p < .001$ ($\eta^2 = .27$), and Perceived Financial Health, $F(6, 330) = 84.73$, $p < .001$ ($\eta^2 = .61$). Post-hoc Tukey tests reveal extensive pairwise differences between clusters; after correction for multiple comparisons, approximately 52–90% of pairwise contrasts remain significant (adjusted $p < .05$). Full ANOVA tables are reported in Supplemental Table S2.

To further assess reducibility, we compare regression models including (a) cluster membership only, (b) socio-demographic variables only, and (c) both predictors jointly. Across outcomes, cluster membership accounts for substantially more variance than socio-demographic variables alone ($R^2 = .19-.61$ vs. $.09-.16$). In combined models, clusters contribute large additional explanatory power beyond SES ($\Delta R^2 = .14-.45$; Figure 5), whereas SES adds comparatively little unique variance once cluster membership is included.

The observed associations between cluster membership and the external outcomes provide evidence that the learned profiles capture underlying behavioral–psychological organization that generalizes across distinct but interconnected domains of functioning, rather than reflecting redundancy among outcome measures.

4.8 Assigning New Individuals to Learned Profiles

In the applied settings that motivated the development of this framework--such as financial advising, behavioral coaching, and clinical contexts--new individuals must be assigned to previously constructed behavioral profiles. It is obviously infeasible to re-cluster from scratch every time a new respondent is added to the population. Accordingly, we evaluate whether the new respondents can be assigned to existing clusters using a supervised classification approach. To this end, we train a supervised ensemble soft-voting classifier (see Supplemental Table C4),

composed of a Support Vector Machine (SVM) and a Random Forest, with cluster indices as target labels, and evaluate its performance using 5-fold stratified cross-validation. The ensemble achieves an accuracy of 79.2%, which substantially exceeds chance-level performance given the number of clusters.

4.9 Anecdotal Success Evidence of Tailored Interventions

In addition to quantitative validation, we collect qualitative feedback from a small number of pilot advisors who used the behavioral profiles revealed by the proposed pipeline in live client engagements. Advisors describe using the profiles to anticipate client behavioral tendencies and inform advisor-mediated interventions, particularly in time-constrained or emotionally salient contexts.

Across examples, advisors report that the profiles are perceived as useful for anticipating patterns of financial avoidance under high salience, asymmetric pacing needs within couples, sensitivity to pressure or authority, and reluctance to request assistance. These anticipated tendencies inform decisions about communication framing, pacing and staging of recommendations, meeting structure, and the timing of information delivery (e.g., asynchronous plan review prior to discussion). Advisors further note that the profiles support interpretation of client behavior when unexpected disclosures or actions occurred, reducing surprise and enabling calibrated responses.

5. Use Case 2: Behavioral Segmentation At Scale

Here we evaluate whether the same two-stage framework identifies meaningful behavioral patterns in a large, independent, population-based dataset with different measurement properties. We apply the framework to the 2017 U.S. CFPB National Financial Well-Being Survey, a nationally representative dataset designed to assess financial well-being, financial behaviors, and

perceived financial capability among U.S. adults. Specifically, we apply the proposed framework to segment the dataset population into meaningful clusters reflecting distinct behavioral profiles.

5.1 Dataset Description and Construct Mapping

Although the CFPB survey contains a broad set of variables spanning demographics, family context, financial behaviors, and structural conditions, our analysis focuses on a targeted subset of items reflecting psychometric, attitudinal, relational, or cognitive constructs relevant to financial decision-making. Variables capturing situational constraints, access-related factors, or external resources are excluded because they do not represent psychological attributes and are therefore not suitable inputs for trait-based behavioral profiling (see Appendix A for item details).

We group the selected items into five domains--Capabilities, Financial Literacy, Planning, Budgeting & Saving, and Social & Relational Support (as shown in Appendix A)--which serve as functional analogues to broader conceptual categories represented in the proprietary dataset.

5.2 Characteristics and Measurement Properties of the Sample

After excluding incomplete cases ($n = 497$; $<8\%$ of respondents), the final analytic sample consists of 5,897 individuals. Incomplete cases were defined as respondents with missing data on one or more items required for domain construction and clustering. The large sample size and broad demographic coverage provide a stringent test of the framework utility under reduced construct resolution and heterogeneous item scales. Although the CFPB survey was not designed as a unified psychometric battery, the selected domains exhibit sufficient variability and internal coherence to support nonlinear embedding and clustering (Cronbach's $\alpha = .78-.87$; Supplemental Table S1). The Financial Literacy domain contains items that have binary response values corresponding to correct or incorrect answers. The remaining domains capture perceived

capability, planning tendencies, routine financial management, and social support.

5.3 Producing Domain Scores

Domain-level scores are derived using the same general principles described in Section 3, with adaptations reflecting the CFPB instrument's structure. Financial Literacy is scored as a difficulty-adjusted accuracy composite in the exact same way as the corresponding proprietary domain. The Social & Relational Support domain, which consists of seven binary items, is scored using a simple average⁴. For the remaining psychometric domains, diagnostic analyses indicate the need for nonlinear embedding, yet linear anchor-based projection in the resulting embedded space is sufficient. Across these domains, UMAP embeddings achieve high trustworthiness (0.944–0.983) and continuity (0.979–0.993); selected hyperparameter values are listed in Supplemental Table S1.

5.4 Clustering and Internal Validation

Following the framework methodology, we tune hyperparameters for the second-stage UMAP and spectral clustering via randomized search, optimizing the same metrics used for the proprietary dataset (Supplemental Table S1). As expected, the Social & Relational Support domain is not suitable for clustering, as it has only eight unique score values across all participants; this domain is therefore excluded from the clustering step and retained solely for post hoc interpretation. More unexpectedly, the inclusion of the Financial Literacy domain consistently collapses the learned manifold, resulting in poor cluster separation and reduced interpretability. Careful inspection of the resulting score distribution reveals that, despite its high nominal response resolution, empirically, Financial Literacy concentrates respondents on a

⁴ The items in this domain appear to be positively correlated in a fairly uniform way, so the covariance weights end up close to equal weighting. When that happens, the covariance-weighted score collapses to something very close to the simple average.

limited set of score values, with the four most frequent values alone comprising approximately 30% of the sample (Figure 4). These spikes impose a strong ordering structure that disproportionately shapes the embedding geometry and obscures multidimensional behavioral variation contributed by other domains. Accordingly, Financial Literacy is also excluded from clustering and examined only post hoc for profile interpretation.

After removing these two domains, we perform the clustering procedure to group the participants into seven clusters (Figure 1c). Internal validation metrics indicate meaningful clustering structure, with a Silhouette Score of 0.500, a Calinski–Harabasz Index of 10,639.98, and a Davies–Bouldin Index of 0.622. Although cluster separation is less pronounced than in the proprietary dataset, these values are consistent with coherent profile differentiation given the larger population base and reduced construct resolution.

As in the case of the proprietary dataset, we assess the seed stability by rerunning the clustering pipeline across 100 independent random seeds and subsample stability by clustering 100 random 80% subsamples within a fixed second-stage embedding. Both metrics indicate that the recovered structure is not driven by stochastic variation or sampling idiosyncrasies: the seed stability is high, with mean ARI = 0.814 (SD = 0.086) and subsample stability is very high, with mean ARI = 0.994 (SD = 0.003)

5.5 Cluster Interpretability

The CFPB clusters exhibit coherent and interpretable structure, reflecting distinct configurations of financial capability, planning, budgeting and saving behavior, social and relational functioning, and financial literacy (Figure 3b). Although cluster means generally increase along an overall gradient of financial functioning, the profiles are not reducible to a single severity dimension. Instead, clusters are distinguished by how knowledge, planning capacity, and enacted financial

behaviors align—or fail to align—within individuals.

Lower-functioning clusters are characterized by comparatively weaker budgeting and saving behavior and planning capacity, even when financial literacy is moderate, suggesting barriers in translating knowledge into sustained financial routines. Intermediate clusters diverge in the balance between structural behaviors and cognitive resources: some profiles emphasize strong budgeting and saving practices with only moderate planning or capability, whereas others show elevated planning and literacy without commensurate behavioral execution. At higher levels of functioning, clusters differentiate based on the relative emphasis placed on planning intensity, behavioral consistency, and social–relational support, distinguishing profiles marked by comprehensive, integrated financial engagement from those with more selectively developed strengths (Table 5b).

5.6 Demographic Analysis

Given the central role of socio-demographic variables in population-based financial surveys, we examine whether cluster membership in the CFPB sample is primarily explained by socio-demographic factors. We estimate a multinomial logistic regression predicting cluster membership from age, education, and income. The overall model is statistically significant (likelihood-ratio test, $p < .001$), and age, education, and income are each statistically significant predictors across multinomial contrasts (all $ps < .001$). However, explanatory power is modest (McFadden pseudo- $R^2 = .061$), indicating that socio-demographic variables account for only a limited proportion of variance in cluster assignment. These results mirror the proprietary dataset findings and indicate that the CFPB clusters are not reducible to demographic stratification, but instead capture behavioral and psychological variation beyond standard SES indicators.

5.7 External Associations With Independent Outcomes

We consider three outcome domains not included in the clustering process: Life Satisfaction and Perceived Financial Health, which parallel the evaluative outcomes examined in the proprietary dataset, and Material Hardship capturing acute financial strain. Cluster membership is significantly associated with all three outcomes. One-way ANOVAs indicate between-cluster differences for Life Satisfaction ($\eta^2 = .14$), Material Hardship ($\eta^2 = .23$), and Perceived Financial Health ($\eta^2 = .44$), closely paralleling the pattern observed in the proprietary data (see Supplemental Table S2). Post-hoc Tukey tests indicate that these effects reflect systematic differences across multiple clusters rather than being driven by a single extreme group.

Further, as shown in Figure 5, cluster membership explains substantial variance in Life Satisfaction ($R^2 = .14$) and Perceived Financial Health ($R^2 = .44$), exceeding the variance explained by demographics alone ($R^2 = .04$ and $.25$, respectively), and contributes meaningful unique variance in the full models ($\Delta R^2 = .12$ for Life Satisfaction; $\Delta R^2 = .25$ for Perceived Financial Health). As expected, socio-demographic variables account for a larger share of variance in Material Hardship ($R^2 = .27$); nevertheless, cluster membership explains additional unique variance ($\Delta R^2 = .09$), indicating that behavioral–psychological profiles provide explanatory power even for outcomes reflecting concrete economic strain.

6. Structural Alignment of Cluster Profiles Across Datasets

We have seen that the proposed framework can be applied to two datasets of very different nature, producing meaningful interpretable clusters in each case. We next explore whether profiles derived from the proprietary and CFPB samples exhibit functional similarities. The challenge arises because the instruments in both datasets are not the same: they contain different questions, with markedly different measurement properties. Still, both instruments assess several core constructs related to financial behavior, literacy, psychological resources, positive

functioning, and planning or organizational processes. Thus, we *harmonize* the two datasets into a common five-domain conceptual space: Financial Behavior, Financial Literacy, Psychological Resources, Positive Functioning, and Motivation & Planning. Financial Construal Level domain is excluded from the proprietary side because the CFPB dataset lacks an equivalent domain. See Appendix D for details on constructing the harmonized space.

This unified space allows direct comparison of profiles derived from both datasets, with alignment quantified using several complementary metrics described in Appendix D2. Cluster centroids are closely aligned along a dominant global functioning axis defined primarily by Financial Behavior, Psychological Resources, and Motivation & Planning (PC1 loadings = .524, .505, and .507, respectively), with weaker contributions from Financial Literacy and Positive Functioning (loadings = .250 and .389). Cluster ordering along this axis is perfectly preserved across datasets (Spearman $\rho = 1.00$), indicating preserved rank structure despite differences in domain coverage. Geometric alignment of cluster profiles is moderate to strong (Procrustes disparity = .158; mean CPSI = .739), with the greatest correspondence observed at higher levels of functioning and increased compression in CFPB profiles relative to the proprietary dataset at the lower end of the distribution. Figure 6 illustrates that, when clusters are ordered along the global functioning axis, proprietary and CFPB centroids exhibit similar absolute functioning levels, consistent with strong ordinal correspondence and relatively low Procrustes disparity.

These results indicate that the framework captures a shared latent structure that extends beyond sample-specific clustering solutions. While the present analysis is limited to two datasets, the observed structural alignment suggests that sufficiently rich behavioral surveys may reveal profiles reflecting inherent properties of financial behavior. Further experimentation with additional datasets—across different instruments and populations—is required to confirm this

conjecture.

7. Discussion

This study presents a novel behavioral segmentation framework that leverages unsupervised learning to uncover latent behavioral structure. The primary contribution of this work is methodological: the framework that introduces a staged approach to behavioral profiling, which separates within-domain representation from cross-domain integration, enabling psychologically interpretable segmentation across diverse psychometric instruments. Rather than assuming a predefined structure or rigid scoring strategy, this approach accommodates domains with varying measurement properties, including both approximately linear and nonlinear relationships among items.

We provide evidence of wide applicability of the proposed framework by using it to uncover latent profiles in two independent datasets of very different nature. The emergent profiles in both cases demonstrate robust internal structure and meaningful differentiation with respect to external outcomes—most strongly for financial health and psychological well-being, and to a lesser degree for life satisfaction. This pattern aligns with prior work showing that proximal outcomes such as financial health are more tightly linked to daily behaviors and psychological capacities, whereas life satisfaction reflects broader and more diffuse life evaluations (Diener et al., 1999; Netemeyer et al., 2018). Systematic differences on these broader outcome measures suggest that the learned profiles reflect underlying psychological organization rather than merely differences in observable financial behaviors.

In addition to quantitative validation above, the framework is being evaluated in live advisory contexts, where preliminary practitioner feedback indicates that the resulting profiles predict behaviorally relevant tendencies that inform practitioners' decision-making in tailoring

advisor–client interactions.

7.1 Methodological Advances

This work advances behavioral segmentation methodology in four key ways. First, rather than clustering item-level responses directly, the framework adopts a two-stage representation strategy that separates the inference of the within-domain structure from cross-domain integration. This design preserves domain-specific psychological distinctions while preventing domains with greater item counts from disproportionately influencing the clustering solution.

Second, the framework formalizes domain-level representations as continuous scores that place individuals along a meaningful psychological axis within the domain. Specifically, we introduce an anchor-based projection strategy that can be applied in either the original item space or after dimensionality reduction, mapping responses onto interpretable continua defined by fixed conceptual reference profiles.

Third, the framework is intentionally modular and algorithm-agnostic. Although UMAP and spectral clustering provide strong empirical performance in the present study, the architecture does not depend on a specific embedding or clustering method, enabling applicability across instruments, populations, and survey formats.

Finally, by applying the same framework to datasets with markedly different psychometric depth, this work demonstrates how measurement richness influences the granularity of recovered behavioral profiles without requiring fundamentally different modeling strategies.

Together, these advances provide a flexible and interpretable approach to behavioral-finance segmentation that extends beyond existing linear or single-pass clustering methods. Domain construction, scoring methods, and inclusion in clustering are specified

conceptually a-priori, and then evaluated *empirically* using the diagnostics described in this paper, leading to potential revision cycles. This enables the framework to adapt flexibly across instruments and datasets while preserving interpretability and methodological coherence.

7.2 Practical Applications

The clustering framework developed in this study lends itself to a wide range of applications across clinical, advisory, and digital financial contexts.

In *clinical psychology and financial therapy*, behavioral profiles may support more targeted intervention planning than traditional demographic or symptom-based approaches by aligning treatment strategies with clients' emotional, cognitive, and motivational readiness for financial change (Archuleta et al., 2012). For example, profiles characterized by strong capability and stable behavioral routines (e.g., Cluster 3) may benefit less from motivational enhancement strategies and more from structured goal-setting, cognitive restructuring around long-term planning, and values clarification interventions. In such cases, therapeutic work may focus on optimizing existing strengths rather than increasing behavioral activation. In contrast, individuals in Cluster 0, who exhibit lower psychological resources or signs of emotional depletion, may benefit from approaches that prioritize therapeutic alliance, emotion-focused techniques, or motivational interviewing (MI) to strengthen autonomy, hope, and readiness for change before introducing structured financial planning tasks. In this way, the profiles may assist practitioners in determining whether cognitive-behavioral techniques, motivational enhancement, or supportive-expressive approaches are most appropriate, thereby facilitating more precise treatment matching within established clinical frameworks.

In *financial advising*, the profiles support personalization of communication style, planning horizons, and behavioral scaffolds based on clients' cognitive and motivational

tendencies. For instance, clients with strong analytical capacity but weak follow-through may benefit from structured action plans and accountability mechanisms, whereas those with lower confidence or elevated anxiety may respond better to incremental goal-setting and simplified messaging.

In *consumer-facing fintech* platforms, behavioral profiles can drive adaptive personalization features such as goal tracking, contextual nudges, and mindset-specific content (Thaler & Sunstein, 2008). Integration with LLM-powered agents further enables real-time adjustment of conversational tone and support strategies based on users' behavioral tendencies. This approach is currently being operationalized within the Calibrate platform, where a reduced-item classifier enables scalable psychometric segmentation to support adaptive onboarding, behavioral coaching, and digital financial planning.

At a *population level*, behavioral segmentation can inform public financial-wellness initiatives and workplace financial education programs, enabling stratified content delivery, early risk identification, and targeted outreach to underserved or vulnerable groups.

7.3 Limitations and Future Work

This study has several limitations that primarily reflect data availability and the scope of empirical evaluation rather than the proposed methodology itself.

First, both datasets are cross-sectional, which precludes assessing how individuals' behavioral profiles evolve over time. It is important to distinguish this limitation from methodological stability: while the clustering solutions are reproducible across random initializations and analytic settings, the present data do not permit evaluating whether individuals transition between profiles in response to interventions, life events, or changing financial circumstances. Longitudinal data would be required to study such dynamics and to examine how

the framework responds to temporal change.

Second, all measures are based on self-report. Self-reported data may be influenced by response styles, social desirability, or differential self-awareness, which can introduce measurement noise and affect the geometry of the learned representations. Incorporating behavioral, transactional, or passively collected data in future work would strengthen ecological validity and improve the precision with which behavioral structure is inferred.

Third, the proprietary dataset is modest in size and drawn from a heterogeneous convenience sample that skews younger than the broader U.S. population. While this reflects a limitation in terms of population representativeness, it also mirrors the data constraints commonly encountered in applied and advisory settings, where large probabilistic samples are rarely available. Demonstrating that the framework yields coherent and interpretable structure in such a small-sample context is therefore a practical feature of the approach. Nonetheless, additional applications across diverse populations, institutional settings, and sample sizes would help further characterize the robustness and boundary conditions of the method.

Fourth, as with most unsupervised learning frameworks, the proposed approach involves analytic design choices--such as domain construction, embedding parameters, and clustering resolution--that can influence the resulting structure. Rather than emphasizing uniqueness of any single solution, we focus on reproducibility across reasonable specifications. Future work could further formalize guidelines for parameter selection and sensitivity analysis to support broader applied use.

Finally, although the present study focuses on financial behavior, the analytic framework itself is domain-agnostic. Applying the approach to other areas--such as mental health, education, or wellness--would help assess the extent to which similar forms of latent

behavioral-psychological structure emerge across different substantive domains and data-generating contexts.

8. Conclusion

This paper introduces a two-stage unsupervised learning framework for identifying psychologically interpretable patterns of financial behavior. Applied to both a proprietary psychometric instrument and the public CFPB dataset, the approach yields coherent and externally meaningful clustering structure, demonstrating its utility for uncovering latent behavioral organization in financial data. Although additional validation across more diverse populations will further clarify its generalizability, the present results show that integrating behavioral science with manifold learning can support theoretically informed and practically useful segmentation of financial mindsets.

We validate the framework through internal measures of psychometric coherence and external associations with independent well-being outcomes, indicating that the recovered structure reflects psychologically relevant variation rather than statistical artifacts. The resulting clusters extend traditional financial personas by incorporating emotional, cognitive, and motivational dimensions, yielding segments that are interpretable and informative in applied settings.

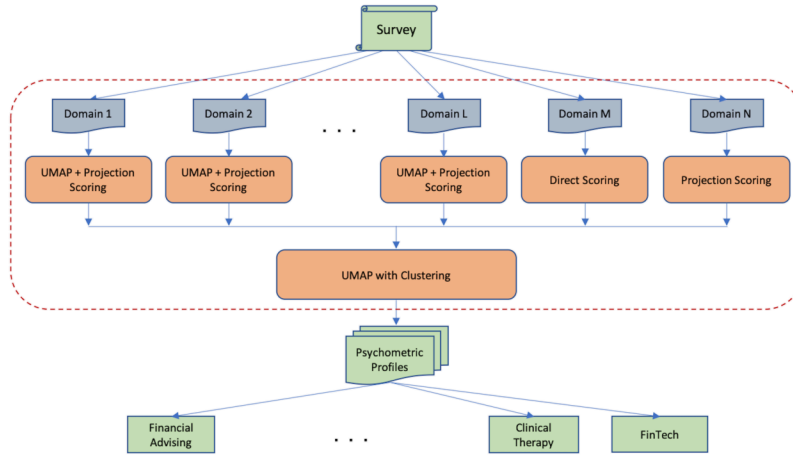
Together, these contributions support a growing movement toward behaviorally intelligent personalization in financial services. The proposed framework offers a scalable and replicable foundation for modeling the human dimensions of financial decision-making in a manner that is data-driven, flexible, and person-centered.

Acknowledgements

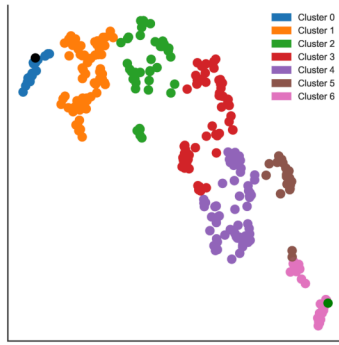
We are grateful to Dr. Joseph Beyer of Microsoft and Dr. Megan McCoy of Kansas State

University for their early guidance and encouragement in the development of the assessment that produced our proprietary dataset. We also thank Dr. McCoy and Dr. Jack Bowler of the University of Utah for their careful reading of the manuscript and valuable feedback.

(a)



(b)



(c)

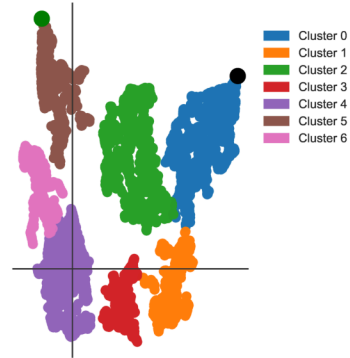


Figure 1. Behavioral-financial clusters derived from unsupervised learning. (a) Schematic of the two-stage unsupervised learning framework. (b) Two-dimensional UMAP projection of the proprietary dataset colored by Spectral Clustering assignments. (c) Two-dimensional UMAP projection of the public CFPB dataset. In both projections, anchor references (black and green dots) are included to aid spatial interpretability of the embedded profiles. The anchors are generated by setting all domain scores to their observed minimum or maximum values across all participants and subsequently embedded into the final space. Cluster structure exhibits clear separation in reduced-dimensional psychometric space, with curvature reflecting dataset-specific nonlinear organization.

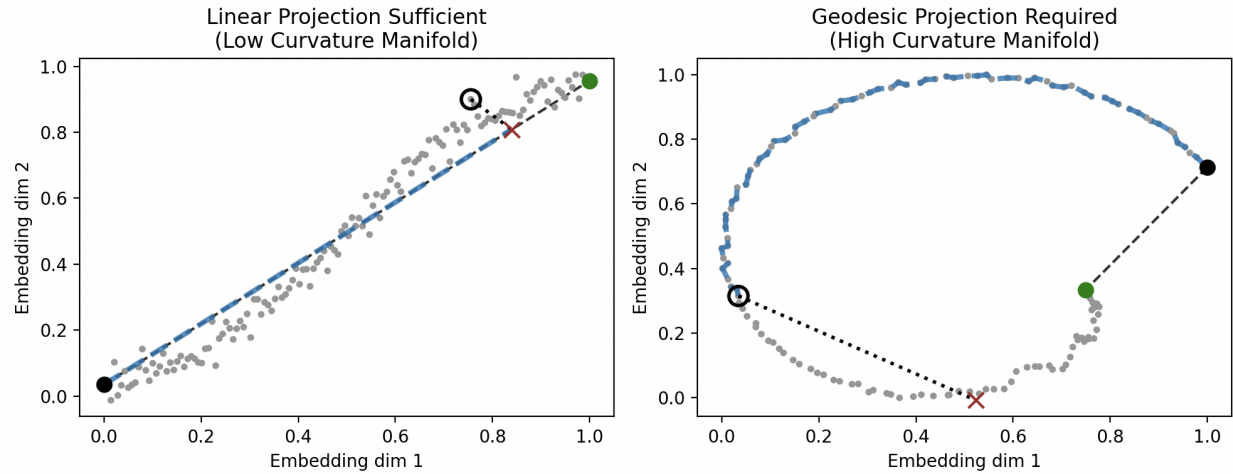


Figure 2. Anchor-based linear versus geodesic scoring on low- and high-curvature manifolds.

Panel A illustrates a domain embedding with low curvature, where participant embeddings (gray points) are organized along an approximately linear latent gradient between the embedded minimum ($v_{\min}^e$, black) and maximum ($v_{\max}^e$, green) anchors. For a representative participant (circled), the orthogonal projection onto the anchor-defined axis yields a scalar score corresponding to the participant's progress along the domain continuum. The blue dashed segment along the anchor axis indicates the resulting linear score.

Panel B illustrates a domain embedding with substantial curvature. Although a straight-line anchor axis can still be defined, linear projection no longer faithfully represents progression along the latent gradient. Instead, the participant's score is defined by the length of the manifold path from $v_{\min}^e$ to the participant along the embedded manifold (blue dashed path), normalized by the total path length between $v_{\min}^e$ (black) and $v_{\max}^e$ (green).

In both cases, the score reflects progression along the anchor-defined domain continuum rather than overall Euclidean proximity to the anchors.

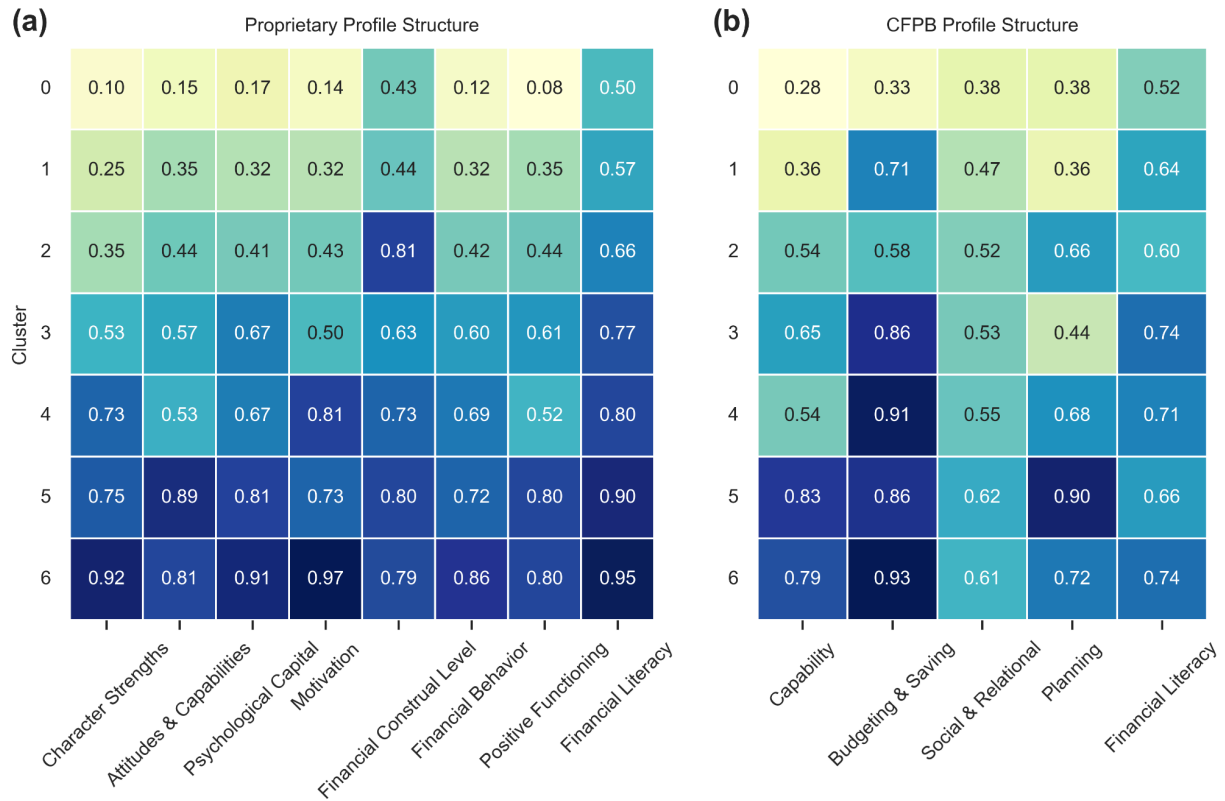


Figure 3. Heatmaps of mean domain scores by cluster across eight psychological and behavioral domains for the proprietary dataset (a) and five domains for the CFPB dataset (b), revealing distinctive cluster-level profiles.

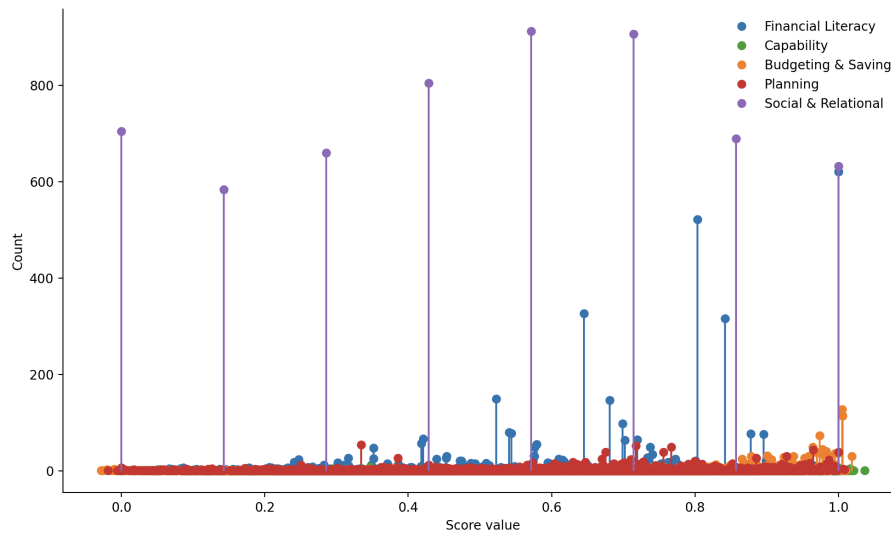


Figure 4. The spike chart of score value occurrences for the CFPB domains. The Social and Relational domain shows spikes corresponding to the eight possible score values. The Financial Literacy domain shows a small number of spikes dominating the score distribution. The rest of the domains show wider distribution of scores across participants, with less prominent spikes.

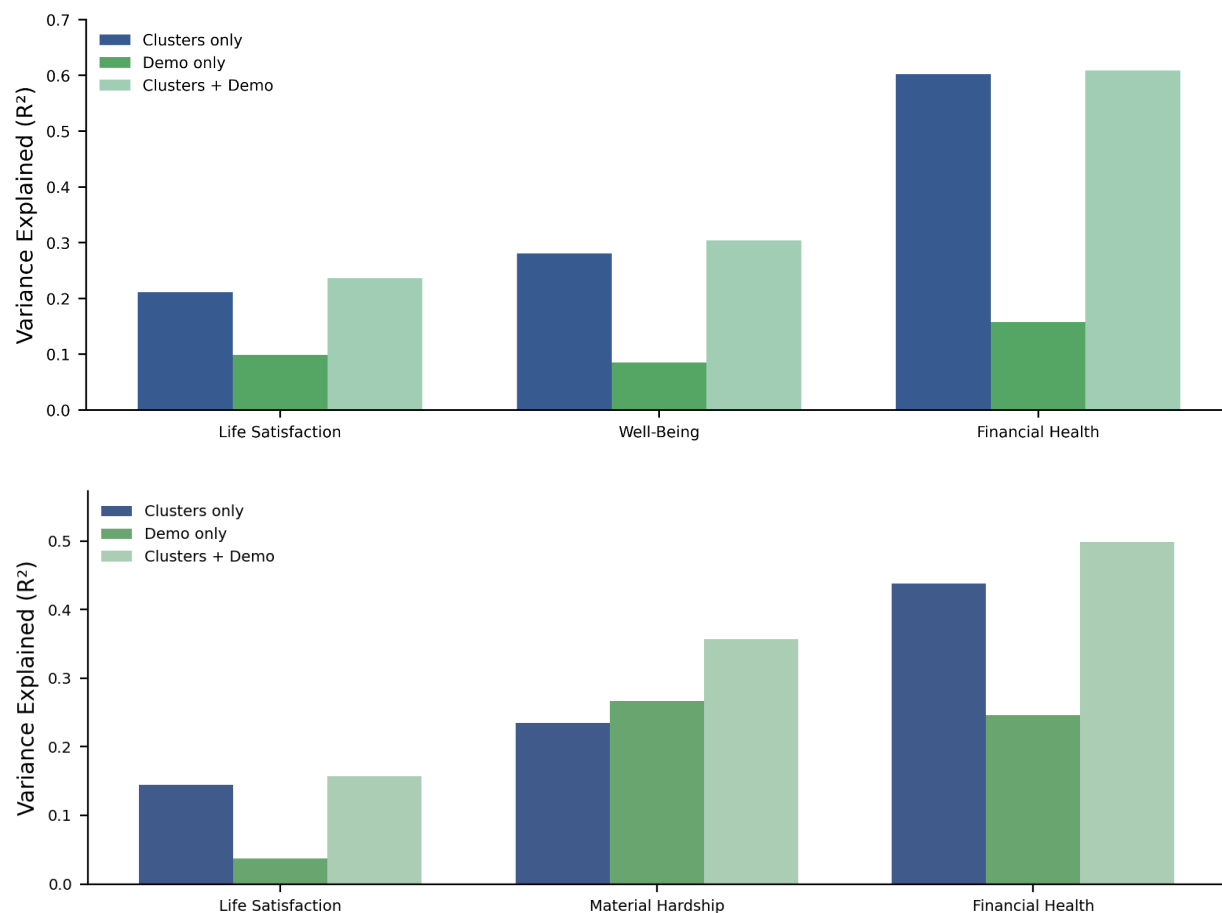


Figure 5. Variance explained by cluster membership and socio-economic status for subjective and financial outcomes in the proprietary (top) and CFPB (bottom) datasets.

Bar charts show the proportion of variance explained (R^2) in Life Satisfaction, Psychological Well-Being, Financial Health, and Financial Strain for three models: (1) *Clusters only*; (2) *Demographics only* (age, education, income); and (3) *Clusters + Demographics*. In both datasets, cluster membership accounts for substantial variance in Life Satisfaction and Financial Health, exceeding that explained by demographics alone. For Material Hardship--a construct more tightly linked to resource constraints--demographics contribute more variance, yet clusters still explain sizable unique variance beyond demographics. The combined models indicate that behavioral-psychological profiles captured by the clustering framework provide explanatory power that is not reducible to demographic factors.

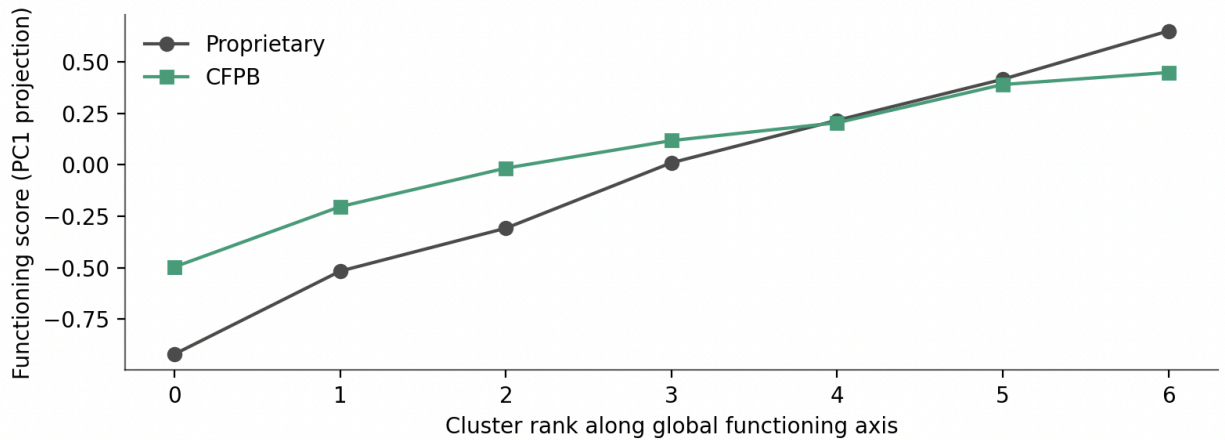


Figure 6: Global functioning gradient across clusters. Clusters from the proprietary and CFPB datasets are ordered by their position along the global functioning axis (PC1), derived from principal component analysis of pooled cluster centroids expressed in the unified five-domain space. Lines connect order-preserving paired clusters to illustrate similarity in absolute functioning levels across datasets.

Table 1

Summary of the eight psychometric domains used in the clustering analysis, including theoretical foundations, item counts, and response formats. Full descriptive statistics and reliability estimates are reported in Supplemental Table S1; sample items are provided in Appendix A.

Domain	Description	Format	Item Count	Core Constructs
Character Strengths	Adapted from the VIA Classification of Character Strengths (Peterson & Seligman, 2004) to reflect financial decision-making contexts	5-point Likert	21	Self-regulation, perseverance, ethical values
Attitudes & Capabilities	Perceived behavioral control and financial self-efficacy; aligns with Family Financial Socialization Theory (Gudmunson & Danes, 2011)	5-point Likert	17	Financial self-efficacy, perceived control
Psychological Capital	Positive psychological resources for coping and adaptive financial behavior, drawn from the PsyCap framework (Luthans et al., 2007)	5-point Likert	8	Hope, resilience, optimism, self-efficacy
Motivation	Internal and external motivational drivers of financial behavior; based on Self-Determination Theory (Ryan & Deci, 2017)	5-point Likert	14	Goal orientation, long-term motivation
Financial Behavior	Everyday financial management behaviors; consistent with stages-of-change perspectives (Prochaska & DiClemente, 1983)	5-point Likert	11	Behavioral consistency, habit strength

Positive Functioning	Psychological well-being and emotional balance relevant to financial decision-making (Seligman, 2011)	5-point Likert	9	Subjective well-being, emotional regulation
Financial Construal Level	Tendency to interpret financial situations at more concrete vs. more abstract levels of meaning (Trope & Liberman, 2010).	Forced choice (2-option)	10	Construal-level processing, abstraction orientation
Financial Literacy	Objective financial knowledge assessed using validated items (Lusardi & Mitchell, 2011)	Multiple choice	5	Decision accuracy, factual financial knowledge

Note: All domain items are derived from validated psychometric frameworks (e.g., VIA Character Strengths, PsyCap, Self-Determination Theory, Construal Level Theory) and are adapted where appropriate to reflect financial decision-making contexts. The final 95 items represent the empirically selected subset of a broader 224-item domain pool, retained based on distributional quality, non-redundancy, and their contribution to stable structure in preliminary UMAP manifold assessments.

Table 2

Summary of Domain Structures, Dimensional Properties, and Scoring Methods for the Proprietary and CFPB Datasets

Domain	# Items	PC1 Variance Explained	Domain Resolution	Nonlinearity Indicator*	Spearman ρ /Kendal τ	Scoring Method
Proprietary Dataset						
Character Strengths	21	.434	4.77×10^{14}	.614	.997/.952	UMAP + Anchor
Attitudes & Capabilities	17	.407	7.63×10^{11}	.588	.994/.932	UMAP + Anchor
Psych. Capital	8	.537	3.91×10^5	.597	.994/.932	UMAP + Anchor
Motivation	14	.465	6.10×10^9	.541	.995/.941	UMAP + Anchor
Financial Behavior	11	.430	4.88×10^7	.573	.996/.951	UMAP + Anchor
Positive Functioning	9	.517	1.95×10^6	.574	.989/.916	UMAP + Anchor
Construal Level	10	.181	1,024	.381		Weighted average
Financial Literacy	5	-	32	-		Difficulty-weighted average, interpretation only
CFPB Dataset						
Capability	13	.424	7.24×10^8	.400	.999/.978	UMAP + Anchor
Budgeting & Saving	9	.346	1.35×10^6	.377	.979/.879	UMAP + Anchor
Planning	9	.362	7.76×10^5	.399	.994/.933	UMAP + Anchor
Financial	12	.123	4,096	.319		Difficulty-weighted

Literacy					average
Social & Relational	7	-	128	-	Simple average, interpretation only

Note: * Nonlinearity Indicator is defined as the mean absolute cosine similarity between the global PC1 direction and the local 1-D PCA tangent estimated from each point's 20 nearest neighbors. Lower values indicate greater deviation from global linearity and stronger support for a nonlinear embedding. Even when values suggest approximate linear structure, incomplete local tangent alignment motivates the use of manifold embedding to preserve curved latent organization.

Domains with very limited response resolution, which yield only a small number of unique response patterns after deduplication, are excluded from PCA- and manifold-based linearity diagnostics because such metrics are not meaningful for discrete, low-variability constructs.

Outcome domains (Life Satisfaction, Psychological Well-Being, Material Hardship, and Financial Health) are not included in these diagnostics because they are used solely as external validation measures rather than as inputs to the embedding or clustering procedures.

Table 3

Demographic characteristics of proprietary set participants

Variable	Category	N	%
Gender	Female	170	50.4%
	Male	147	43.7%
	Non-binary/undisclosed	20	5.9%
Age Group	18–24	86	25.5%
	25–34	41	12.2%
	35–44	85	25.2%
	45–54	45	13.4%
	55–64	48	13.9%
	65+	18	5.3%
	Not reported/undisclosed	15	4.5%
Income	Less than \$50,000	77	22.9%
	\$50,000 – \$100,000	50	14.8%
	\$100,000 – \$200,000	74	22.0%
	\$200,000 – \$500,000	69	20.5%
	\$500,000 or more	42	12.5%
	Not reported	25	7.4%
Education	Graduate or professional degree	115	34.1%
	Bachelor's degree	115	34.1%

	High school or equivalent	24	7.1%
	Some college / Associate degree	39	11.6%
	Not reported/Undisclosed	44	13.1%
Race/Ethnicity	White	202	59.9%
	Black or African American	51	15.1%
	Asian	42	12.5%
	Hispanic or Latino	19	5.6%
	Other	19	5.6%
	Not reported	6	1.8%

Note: Participants could select multiple ethnicity categories; percentages sum to more than 100%.

Table 4

Cluster distribution

Both datasets exhibited similar cluster dispersion: cluster sizes ranged from 7%–21% in the proprietary sample and 8%–22% in the CFPB sample.

Proprietary Dataset			CFPB dataset		
Cluster	n	% of Sample	Cluster	n	% of Sample
0	26	7.72%	0	1,301	22.06%
1	71	21.07%	1	745	12.63%
2	47	13.95%	2	1,294	21.95
3	65	19.29%	3	980	16.62
4	62	18.40%	4	461	7.82
5	25	7.42%	5	629	10.67
6	41	12.17%	6	487	8.26
<i>Total</i>	337	100	<i>Total</i>	5,897	100

Table 5a

Descriptive Cluster Profiles in the proprietary Dataset

Cluster-level summary descriptions derived from domain score patterns (Character Strengths, Attitudes & Capabilities, Psychological Capital, Motivation, Financial Behavior, Positive Functioning, Financial Construal Level, and Financial Literacy). Labels are descriptive and reflect relative strengths across domains.

Cluster	Label	Interpretive Summary
----------------	--------------	-----------------------------

0	Low capability, low psychological resources	Globally low strengths, capability, psychological capital, motivation, and behavior. Adequate literacy does not translate into action because confidence and functioning are limited.
1	Emerging capability with inconsistent habits	Improving capability, literacy, and functioning, but motivation and routines are not yet stable. Behavior is inconsistent despite modest psychological resources.
2	Cognitively engaged, moderately supported	Moderate strengths, capability, and behavior, with relatively higher literacy and financial construal. Thinks about finances and plans ahead, but only mid-level energy and execution.
3	Capable and stable	Strong capability, literacy, functioning, and behavior, paired with moderate motivation. Operates effectively through competence and routine rather than pronounced motivational drive.
4	Highly capable, steady performer	High capability, psychological capital, functioning, and literacy, with solid behavior but only moderate motivational drive. Performs reliably through skills and structure rather than strong emotional push.
5	Organized and well-resourced	High strengths, capability, motivation, and behavior, with very strong literacy and functioning.
6	Highly resourced and consistently effective	Exceptionally strong across all psychological, motivational, and behavioral domains. Demonstrates very high internal resources, strong motivation, excellent functioning, and reliable execution.

Table 5b

Descriptive Cluster Profiles in the CFPB Dataset

Cluster-level summary descriptions derived from domain score patterns (Capability, Budgeting & Saving, Planning, Social & Relational Support, and Financial Literacy). Labels are descriptive and reflect relative strengths across domains.

Cluster	Label	Interpretive Summary
0	Low structure, low follow-through	Has difficulty getting started; weak systems, low motivation, and limited follow-through. Confidence is modest, making financial tasks feel effortful or overwhelming.
1	Growing structure, limited action	Has some organizational footing and emerging confidence, but actions are irregular. Motivation exists but isn't strong enough to sustain habits.
2	Highly organized, cautious executor	Well-organized and deliberate, but often slow to execute. Strong systems, but prefers caution over speed.
3	Strong everyday habits	Shows reliable real-world behavior even without elaborate planning. Confident and motivated enough to maintain consistent routines.
4	Structured and consistent	Combines strong systems with strong execution. Confident, organized, and steady in follow-through.

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Appendix A: Domain Composition for Clustering and Validation

A1. Proprietary Dataset Sample Items

1. “I am confident I can make wise financial decisions even under pressure.”
2. “I can confidently navigate uncertain financial waters.”
3. “I believe I can reach my long-term financial goals through persistent effort.”
4. “I do not give up on my financial goals easily.”
5. “Unexpected expenses totally stress me out.”
6. “I lie awake in bed at night because I am worried about my financial future.”
7. “When I get a bill notification, it feels like a threat or warning.”
8. “I often avoid looking at my checking or credit card statement to avoid seeing how much I’ve spent.”
9. “Actually, I gave up trying to keep track of my income and expenses.”
10. “I enjoy learning about personal financial management.”
11. “I actively seek new information about how I can manage my money effectively.”
12. “I see the big picture when planning my finances.”

A2. CFPB Dataset

Capability:

FS1_1-FS1_6, FS2_1-FS2_3, GOALCONF, SELFCONTROL_1-SELFCONTROL_3

Measures perceived financial skills, confidence, sense of control, and self-regulatory capacity.

Budgeting & Saving:

MANAGE1_1-MANAGE1_4, SAVEHABIT, FRUGALITY, ABSORBSHOCK, ENDSMEET, SCFHORIZON

Captures day-to-day money management practices, saving habits, frugality, and short-term financial resilience.

Planning:

PROPPLAN_1-PROPPLAN_4, ACT1_1, ACT1_2, ASK1_1, ASK1_2, FINGOALS

Assesses planning behaviors, proactive financial organization, and the extent to which individuals set and act on financial goals.

Social & Relational:

FINSOC2_1-FINSOC2_7

Measures access to social and interpersonal support, relational resources, and financial well-being derived from social networks.

Financial Literacy:

FK1correct - FK3correct, KH1correct -KH9correct

Measures financial knowledge, understanding of compound interest, diversification, risk.

LifeSatisfaction:

SWB_1 - SWB_3

Captures global cognitive evaluations of life quality.

Perceived Financial Health:

FWB1_1-FWB1_6, FWB2_1-FWB2_4

Uses standard 10-item Financial Well-Being Scale.

Material Hardship:

MATHARDSHIP_1 - MATHARDSHIP_6, DISTRESS, HOUSING

Captures experiences of material hardship and acute financial stress.

The Social & Relational domain is excluded from the clustering inputs because it exhibits very low structural variability (at most 128 possible combinations), providing broad global gradients rather than fine-grained psychological information. Including such low-variability domains in a nonlinear manifold risks collapsing the embedding onto a single severity axis, overshadowing the richer multi-dimensional variation captured by the core behavioral and psychological domains. For this reason, this domain is retained only for post-hoc interpretation and validation of the final cluster solution.

See [CFPB survey codebook](#) for additional information

Appendix B: Benchmarking Data Representations and Clustering Methods

To develop a stable and psychologically interpretable clustering pipeline, we conducted three complementary analyses. First, we examined the geometric properties of the psychometric domains (Stage 1) to determine whether linear dimensionality reduction (e.g., PCA) could adequately preserve global and local structure, or whether nonlinear embedding was required. Second, we benchmarked alternative low-dimensional representations (raw domain scores, PCA projections, UMAP embeddings) to evaluate their suitability for Stage 2 clustering. Third, within the preferred representation, we compared several clustering approaches to assess their qualitative stability and alignment with known behavioral structure. Together, these analyses provide an empirical foundation for the dimensionality-reduction and clustering choices used in the main framework.

B.1 Benchmarking Linear vs. Nonlinear Representations in Stage 1

To evaluate whether a linear method such as PCA could adequately serve as the Stage 1 dimensionality reduction step, we assessed domain-level geometry using three complementary diagnostics:

1. **Global linearity:** the proportion of variance explained by the first principal component (PC1).
2. **Local tangent alignment:** the mean $|\cos \theta|$ between the domain-level PC1 axis and local one-dimensional PCA directions estimated from each participant's 20-nearest neighbors.
3. **Neighborhood preservation:** trustworthiness and continuity comparisons between PCA and UMAP.

Global Linearity (PC1 Variance): Across domains, PC1 captures only 35–59% of total variance, indicating that no domain is dominated by a single global linear axis. This suggests that Stage 1 representations must account for multi-dimensional or curved manifold structure rather than relying on a single major linear direction.

Local Tangent Alignment: Local tangent–global axis alignment is modest, with mean $|\cos \theta|$ values ranging from 0.38 to 0.67 (median values ≈ 0.37 – 0.74 across domains). These moderate alignments indicate substantial curvature or heterogeneous local variance structures, further suggesting that a strictly linear mapping such as PCA is unable to capture the local geometry that Stage 1 must preserve.

Neighborhood Preservation: PCA with two components shows consistently weaker neighborhood preservation compared with UMAP.

- Trustworthiness: PCA ≈ 0.81 – 0.91 , UMAP ≈ 0.94 – 0.98
- Continuity: PCA ≈ 0.93 – 0.96 , UMAP ≈ 0.96 – 0.99

Even when PCA is expanded to four components, UMAP matches or exceeds it across most domains in both metrics, underscoring that the latent structures underlying these psychometric

domains are meaningfully nonlinear.

Across diagnostics--global variance structure, local tangent geometry, and neighborhood preservation--PCA fails to capture the nonlinear manifold characteristics present in the domain-level data. UMAP provides substantially better preservation of both local and global structure, supporting its use as the Stage 1 embedding method.

B.2 Benchmarking Linear vs. Nonlinear Representations in Stage 2

We evaluated whether nonlinear embedding is necessary for identifying stable clusters by comparing clustering performance across three representations: raw domain scores, linear PCA projections, and UMAP embeddings. Although t-SNE is a popular nonlinear visualization tool, it does not preserve global structure and is not suitable for graph-based clustering; therefore, it was excluded from quantitative comparison. Because the proprietary and CFPB datasets differ in size and sampling density, results are reported separately.

Proprietary dataset: This dataset exhibits relatively strong inherent structure, reflected in higher absolute silhouette values across all representations. Clustering on raw domain scores produced a silhouette of approximately 0.17. PCA projections improved separation modestly (≈ 0.38 in 2D; ≈ 0.20 in 4D). UMAP yielded the clearest manifold structure and the strongest separation (≈ 0.56).

CFPB dataset: In this larger and denser dataset, clustering on raw domain scores again produced weak separation (≈ 0.23), indicating that the cross-domain space is not linearly separable. PCA offered only modest improvement (≈ 0.32). UMAP achieved substantially stronger separation (≈ 0.50) and outperformed linear methods on Calinski–Harabasz and Davies–Bouldin indices, consistent with meaningful nonlinear structure in the data.

Across both datasets, linear representations--either raw space or PCA--are unable to recover the nonlinear geometry required for stable, interpretable clustering. UMAP produced

markedly stronger separability and more coherent manifold structure, supporting its use as the Stage 2 embedding in the proposed framework.

B.3 Algorithmic Evaluation of Clustering Methods

Several clustering algorithms are evaluated on the UMAP embeddings

K-means: Considered for its simplicity and widespread use, K-means consistently underperformed. Its assumption of convex, spherical clusters conflicted with the observed structure of the UMAP space. The resulting clusters lacked behavioral coherence, and internal validation metrics (e.g., silhouette score) are poor.

DBSCAN and HDBSCAN: Density-based methods showed initial promise but proved unstable. DBSCAN often merged distinct subgroups, while HDBSCAN labeled large portions of the data as noise. Both are highly sensitive to parameter settings (e.g., *min_samples*, *min_cluster_size*), and qualitative inspection revealed weak alignment with psychological or behavioral patterns.

Agglomerative Clustering: This method was more stable than density-based approaches and delivered modest performance. However, the absence of a clear criterion for selecting the optimal number of clusters, along with weaker separation metrics, limited its usefulness.

Spectral Clustering: Spectral Clustering demonstrated the strongest performance across both quantitative and qualitative criteria. It is well-suited to capturing non-convex cluster shapes and leverages the pairwise similarity structure preserved in the UMAP embedding.

Appendix C: Dimensionality Reduction, Clustering, and Classification Hyperparameters

Supplemental Table C1

UMAP parameters for domain-level (Stage 1) and composite embeddings (Stage 2)

Dataset	Domain	n_neighbors	n_components	min_dist	spread	local_connectivity
Proprietary	Character Strengths	24	15	.8	1.00	1
	Attitudes & Capabilities	24	15	.8	1.00	1

	Psychological Capital	22	8	.6	1.00	1
	Motivation	26	8	.6	1.00	1
	Financial Behavior	26	8	.6	1.00	1
	Positive Functioning	21	5	.4	1.00	1
	Composite Embedding	15	2	.0	.3	7
CFPB	Capability	37	5	.6	1.00	1
	Budgeting & Saving	16	5	.6	1.00	1
	Planning	17	5	.7	1.00	1
	Composite Embedding	24	2	.02	1.2	7

Supplemental Table C2

Spectral Clustering parameters (Stage 2)

Dataset	n_clusters	affinity	gamma	eigen_solver
Proprietary	7	rbf	8.00	arpack
CFPB	7	rbf	2.53	arpack

Supplemental Table C3

Imputation details

Component	Parameter	Value
IterativeImputer	estimator	RandomForestRegressor
	max_iter	10
	initial_strategy	mean
	imputation_order	ascending
	skip_complete	True
	min_value	0 for binary, 1 for Likert
	max_value	1 for binary, 5 for Likert
RandomForestRegressor	n_estimators	250
	max_depth	15

Supplemental Table C4

Classification details

Component	Parameter	Value
Feature Selection (ExtraTreesClassifier)	n_estimators	1000

Ensemble Classifier (VotingClassifier)	max_depth	None
	voting	soft
	base estimators	RandomForestClassifier, SVC
RandomForestClassifier (model_0)	n_estimators	150
	max_depth	10
	max_features	log2
	min_samples_leaf	2
	criterion	gini
SVC (model_1)	bootstrap	False
	kernel	poly
	degree	2
	C	0.5
	gamma	0.05
	probability	True
	shrinking	False

Appendix D: Harmonization of Domains Across Datasets

D1. Domain Harmonization

To explore cross-dataset alignment, we develop a unified five-domain conceptual space shared by both instruments. The goal of this harmonization is to align instrument-specific domains according to their underlying psychological functions rather than their exact item structure or naming conventions. This approach emphasizes functional equivalence rather than item-level correspondence: each conceptual domain aggregates the set of original domains that, within each instrument, represent similar underlying processes (e.g., agency, planning, execution, or motivation). Below we summarize the domain mappings used for the analysis.

Supplemental Table D1

Domain Harmonization Mapping

Conceptual Domain	Proprietary Domains	CFPB Domains
Financial Behavior	Financial Behavior	Budgeting & Saving

Financial Literacy	Financial Literacy	Financial Literacy
Positive Functioning	Positive Functioning	Social & Relational Support
Psychological Resources	Attitudes & Capabilities; Psychological Capital; Character Strengths	Capability
Motivation & Planning	Motivation	Planning

Note. Mappings reflect correspondence at the level of higher-order psychological constructs rather than item-level equivalence. CFPB Social & Relational Support captures aspects of the broader Positive Functioning domain in the proprietary instrument, and CFPB Planning reflects the behavioral expression of the same organizational tendencies represented in the proprietary Motivation domain. These conceptual parallels allow the two datasets to be harmonized within a shared domain space, even though specific item content differs. No direct CFPB analogue exists for proprietary Financial Construal constructs.

Because the instruments differ in the number and granularity of available scales, some conceptual domains are formed by averaging multiple source domains, while others correspond to a single scale. For conceptual domain D_k composed of n_k source domains s_i , the harmonized score for a

participant is the simple mean: $D_k = \frac{1}{n_k} \sum_{i=1}^{n_k} s_i$. This transformation preserves relative standing on the underlying constructs even when item content differs.

D2. Cross-Dataset Cluster Alignment

Since clusters are identified independently within each dataset using the full set of original domain scores, to compare clusters across datasets, we represent each cluster by a *harmonized cluster profile*: a vector of mean scores on the five harmonized domains. For cluster c containing

N_c individuals and harmonized domain j : $\bar{D}_{c,j} = \frac{1}{N_c} \sum_{n \in c} D_{n,j}$. Each harmonized cluster is

therefore represented by a 5-dimensional profile vector: $\bar{D}_c = (\bar{D}_{c,1}, \dots, \bar{D}_{c,5})$. These vectors constitute the basis for cross-dataset comparison.

We evaluate similarity between clustering solutions using several complementary metrics that capture local proximity, ordinal correspondence, and global geometric alignment. First,

pairwise cluster proximity is summarized using the *Cluster-Pair Similarity Index (CPSI)*, a normalized inverse-distance measure computed from Euclidean distances between five-domain cluster profiles. CPSI rescales distances onto an interpretable similarity scale ranging from 0 to 1, with larger values indicating more similar profiles: $CPSI_{ij} = \frac{1}{1+d(i,j)}$, where $d(i,j)$ denotes the Euclidean distance between cluster i and cluster j in the unified five-domain space. CPSI is used descriptively and does not impose a one-to-one correspondence between clusters.

Second, we derive a shared global functioning axis by applying principal component analysis (PCA) to the pooled set of proprietary and CFPB cluster centroids, each represented as a five-dimensional domain profile. The first principal component (PC1) defines a one-dimensional gradient summarizing dominant variation across clusters in the harmonized domain space. Each cluster receives a score on this axis, and cross-dataset alignment in relative cluster ordering is assessed using the *Spearman rank correlation* between PC1 scores, with higher correlation values indicating stronger agreement in cluster ordering across datasets.

Finally, to assess configuration-level geometric similarity, we apply Procrustes analysis to the proprietary and CFPB centroid matrices expressed in the unified domain space. Procrustes analysis identifies the optimal rotation and scaling that best superimposes one configuration onto the other. Geometric similarity is quantified using the Procrustes disparity, defined as the squared Frobenius norm of the residual matrix after optimal alignment: $D = \|X - Y^*\|^2$, where X denotes the matrix of proprietary cluster centroids (rows = clusters, columns = domains) and Y^* denotes the optimally transformed CFPB centroid matrix. Lower disparity values indicate greater similarity in the overall spatial organization of clusters across datasets.

D3. Interpretation Caveats

It is important to note several interpretation caveats. The harmonized domains represent

conceptual correspondences, not item-level equivalences, and thus should not be interpreted as measuring the same constructs with identical psychometric content. Averaging across contributing scales preserves relative differences, profile shape, and cross-domain patterns, but does not reproduce fine-grained item structure or latent-factor detail. Comparisons therefore reflect similarity in high-level psychological organization, not equality of measurement precision or scale composition. Finally, magnitude differences across datasets (for example, compression of variation in the CFPB domains) are expected given differences in survey breadth and scale design. Alignment metrics are consequently interpreted in terms of pattern, ordering, and overall configuration, which provide more stable and meaningful indicators of structural correspondence across instruments.

Supplemental Table S1

Descriptive statistics for each psychometric domain

Reports the number of items, scale ranges, descriptive statistics, and internal consistency (Cronbach's α) for each domain. All domains exhibit adequate variability and reliability for unsupervised modeling.

Domain	# Items	Scale Range	Mean	SD	Min	Max	Cronbach's α
Proprietary Dataset							
Character Strengths	21	1–5	3.76	1.18	1.00	5.00	0.930
Attitudes & Capabilities	17	1–5	2.74	1.40	1.00	5.00	0.897
Psychological Capital	8	1–5	3.78	1.02	1.00	5.00	0.894
Motivation	14	1–5	2.99	1.47	1.00	5.00	0.823
Financial Behavior	11	1–5	3.25	1.46	1.00	5.00	0.875
Positive Functioning	9	1–5	3.19	1.46	1.00	5.00	0.856
Financial Construal Level	10	0/1	0.65	0.47	0.00	1.00	0.651
Financial Literacy	5	0/1	0.84	0.21	0.00	1.00	0.461
Life Satisfaction	5	1–5	3.33	1.14	1.00	5.00	0.865
Perceived Financial Health	10	1–5	2.94	1.41	1.00	5.00	0.922
Psychological Well-Being	6	1–5	3.07	1.37	1.00	5.00	0.570
CFPB Dataset							
Capabilities	13	1–4/1–5	3.35	0.93	1.00	5.00	0.878

Planning	9	1–5, 0/1	3.25	1.35	1.00	5.00	0.803
Budgeting & Saving	9	1–5/1–6	3.97	1.31	1.00	6.00	0.796
Social & Relational	7	0/1	0.51	0.50	0.00	1.00	0.779
Financial Literacy	12	0/1	0.74	0.44	0.00	1.00	0.688
Life Satisfaction	3	1–7	5.44	1.44	1.00	7.00	0.793
Perceived Financial Health	10	1–5	3.35	1.20	1.00	5.00	0.917
Material Hardship	8	1–3/1–5	2.81	0.59	1.00	5.00	0.754

Note: Financial Literacy items are scored 1 = correct, 0 = incorrect, and the domain score reflects the proportion answered correctly.

Supplemental Table S2. Outcome differentiation by cluster membership.

This table reports the results of one-way ANOVA examining how cluster membership predicts three outcome variables: Life Satisfaction, Psychological Well-Being, Material Hardship, and Perceived Financial Health.

Proprietary=df(6, 330), CFPB=df(6, 5890)

Outcome	SSB	SSW	F	Partial η^2	Levene's W	Levene's p
Proprietary Dataset						
Life Satisfaction	3.146	13.330	12.984	0.191	0.68	0.667
Psychological Well-Being	2.283	6.309	19.904	0.266	1.99	0.067
Perceived Financial Health	11.849	7.691	84.732	0.606	7.03	<.001
CFPB Dataset						
Life Satisfaction	1247.742	7398.907	165.546	0.144	24.72	< .001
Material Hardship	174.600	568.844	301.310	0.234	112.63	< .001
Perceived Financial Health	1996.911	2558.232	766.271	0.438	18.84	< .001

Note. Partial η^2 is reported as effect size. Levene's test p-values assess homogeneity of variance. In the proprietary dataset (N = 337), cluster sizes are reasonably balanced (n = 25–71 per cluster), and observed effect sizes are moderate to large (partial η^2 = .19–.61), supporting reliable estimation of between-cluster differences.